

JD Institute ERP — Progress Report

Prepared for: Client review **Last updated:** June 2026 **Platform:** Web-based (works on desktop, tablet, mobile browsers)

1. Project Overview

The JD Institute ERP is a complete web-based management system that replaces the older PHP-based system. It runs entirely from a browser — staff, students, and parents only need an internet connection. There is nothing to install on individual computers.

The system has two main parts: - **Staff Portal** (admin, counsellors, faculty, HR) — for day-to-day operations. - **Student Portal** — students sign in to a separate, simpler interface to see their own information.

Both portals share the same underlying database, so any update on the staff side is immediately visible to students (no syncing or batch jobs required).

2. Modules Built So Far

The system is organised into modules. Each one covers a complete business process from start to finish.

2.1 Lead Management (CRM)

Manages student enquiries from first contact through to enrolment.

Capability	Status
Capture leads (manual entry + public form + bulk import)	✓ Done
Lead status workflow (New → Follow-up → Hot/Warm/Cold → Application → Enrolled)	✓ Done
Call follow-up logging with outcome categories (Hot / Warm / Cold / Not answering / Not connected / Enrolled)	✓ Done
Outcome disposition tracking (e.g. "Planning to visit campus", "Fees high", "Joined elsewhere")	✓ Done
Mandatory outcome before status change (prevents leads from being moved without notes)	✓ Done
Send application link via Email + SMS in one click	✓ Done
Send fee payment link via Email + SMS	✓ Done
Send welcome email after enrolment	✓ Done
Lead reassignment between counsellors	✓ Done

Capability	Status
Lead pools (group leads for bulk assignment)	✓ Done
Lead status history (full audit trail of every transition)	✓ Done
Overdue hot-lead escalation (auto-notifies manager when a hot lead has no contact for 24 hrs)	✓ Done
Counsellor can close/open the public application form	✓ Done

2.2 Public Application Form

The form prospective students fill out to apply.

Capability	Status
Open application form (accessible without login)	✓ Done
Form locks when counsellor closes it (prevents late submissions)	✓ Done
Direct integration with the CRM (submissions appear as leads)	✓ Done

2.3 Admissions & Student Records

Manages enrolled students and their academic records.

Capability	Status
Student profile (personal details, contact info, education history)	✓ Done
Parent contact information	✓ Done
Document uploads (ID proofs, certificates)	✓ Done
Education history (10th, 12th, previous institutes)	✓ Done
Student remarks (counsellor notes per student)	✓ Done
Enrolment into batches (link student to program + campus + intake)	✓ Done
Institute email + portal credentials auto-generation	✓ Done
"Reset & send credentials" button (emails new password to student)	✓ Done
Send student handbook via email	✓ Done
Batch promotion (move full batch to next semester)	✓ Done
Batch graduation (mark batch as graduated, freeze records)	✓ Done

2.4 Fees & Payments

Tracks what each student owes and what they've paid.

Capability	Status
Fee installment scheduling (per enrolment)	✓ Done
Bulk-create installments across multiple students	✓ Done
Fee receipts (cash, cheque, online)	✓ Done
Receipt cancellation with reason	✓ Done
Printable PDF receipts	✓ Done
Fee concessions (approval workflow)	✓ Done
Concession decisions (approve / reject by authorised staff)	✓ Done
Student-side fee summary (balance, paid, due)	✓ Done
Online fee payment link (sent via Email + SMS)	✓ Done
Fee collection counter screen	✓ Done

2.5 Academics (Timetable, Attendance, Courseware)

Capability	Status
Weekly timetable scheduling	✓ Done
Conflict detection (faculty double-booking, room clashes)	✓ Done
Bulk weekly publish (apply timetable to whole batch)	✓ Done
Take attendance (per slot, per batch)	✓ Done
Attendance roster (mark present / absent / late)	✓ Done
Attendance freeze (lock attendance after a deadline)	✓ Done
Batch attendance reports	✓ Done
Absence notifications to student + parent (Email + SMS)	✓ Done
Courseware library (topics, lecture notes, attachments)	✓ Done
Assignment publishing (with file attachments)	✓ Done
Student assignment submissions	✓ Done
Tests & exam management	✓ Done
Marks entry & test results	✓ Done

2.6 Student Portal

The student-facing app at a separate web address.

Capability	Status
Login / Forgot password / Change password	✓ Done
Dashboard (overview of attendance, upcoming classes, pending assignments)	✓ Done
View attendance calendar + monthly report	✓ Done
View timetable	✓ Done
View assignments + submit work	✓ Done
Access courseware (lecture notes, attachments)	✓ Done
Take tests online	✓ Done
View test results	✓ Done
Leave application (request leave from classes)	✓ Done
Profile + documents view	✓ Done
Feedback submission	✓ Done

2.7 HR — Employees, Leaves, Relieving

Capability	Status
Department + Designation master setup	✓ Done
Employee records (personal, role, campus assignment)	✓ Done
Activate / Deactivate employees	✓ Done
Leave types & yearly allocations	✓ Done
Bulk leave allocations (assign yearly quotas in one go)	✓ Done
Employee leave applications	✓ Done
Leave approval workflow	✓ Done
Resignation / Relieving workflow	✓ Done
Approval, rejection, withdrawal flows	✓ Done
Generates Relieving Letter (PDF)	✓ Done
Generates Experience Letter (PDF)	✓ Done

2.8 Student Leaves

Separate from employee leaves — students can request leave from classes.

Capability	Status
Student leave requests	✓ Done
Counsellor approval/rejection	✓ Done
Status notifications to student	✓ Done

2.9 Tasks

Internal staff task management.

Capability	Status
Create + assign tasks (with deadlines)	✓ Done
Task completion tracking	✓ Done
Email notifications on task assigned + completed	✓ Done

2.10 Notifications System

Every Email and SMS that leaves the system passes through this layer.

Capability	Status
Email delivery (via MSG91 + Gmail SMTP fallback)	✓ Done
SMS delivery (via MSG91 DLT-compliant flow)	✓ Done
Templated messages (28 pre-registered templates so far)	✓ Done
Variable substitution (name, dates, links auto-filled)	✓ Done
Scheduled notifications (fire at a future date)	✓ Done
Dispatch log (every Email/SMS recorded with delivery status)	✓ Done
Provider switching via configuration (no code change needed to swap MSG91 ↔ BulkSMS)	✓ Done

2.11 Administration

Capability	Status
User management (create, edit, deactivate staff users)	✓ Done
Role-based permissions (control what each role can see / do)	✓ Done
Password reset by admin (with auto-email of new password)	✓ Done
Self-service forgot-password flow	✓ Done
Campus assignment per user	✓ Done

Capability	Status
Activity audit log	✓ Done
Audit reports (who did what, when)	✓ Done

2.12 Master Data

Configurable lists that drive the rest of the system.

Capability	Status
Institutes (JD School of Design, JD Institute of Fashion Technology, etc.)	✓ Done
Campuses (per institute)	✓ Done
Programs / Courses	✓ Done
Batches (intake + program + campus)	✓ Done
Subjects	✓ Done
Academic sessions	✓ Done
Lead sources (where leads come from)	✓ Done
Lead status master	✓ Done

3. Third-Party Services Integrated

The system relies on a few external services for specific jobs. These are all standard, widely-used providers chosen for reliability and India-region compliance.

Service	What It's Used For	Status
MSG91 Email	Transactional emails from <code>mail.jdinstitute.com</code> (welcome notes, fee receipts, credentials, password resets)	✓ Connected & sending
MSG91 SMS	DLT-compliant SMS (application links, fee links, attendance alerts) — required by Indian telecom regulations	✓ Connected (templates pending verification)
Gmail SMTP (Workspace)	Fallback email channel when MSG91 has a template that isn't registered yet	✓ Connected

Service	What It's Used For	Status
BulkSMSTGateway	Legacy SMS option (kept available — used in the old PHP system)	✓ Connected (currently disabled)
PythonAnywhere	Hosting provider — runs the application 24x7	✓ Deployed at mahirsodagar.pythonanywhere.com
TinyURL	Auto-shortens long application & fee links so they fit in SMS messages	✓ Connected
TRAI DLT Registry	India's regulatory SMS registry. All our SMS templates use DLT-approved Principal Entity templates (JDEDUC)	✓ Registered
Cloudflare	Sits in front of MSG91. We've configured our requests to pass its bot-detection so messages reliably go through	✓ Handled

4. How the Email & SMS System Works

Because most modules send email or SMS at some point (admission confirmation, fee receipts, leave decisions, attendance alerts, etc.), it's worth understanding the design.

Templates are central. Every email or SMS the system sends is based on a pre-defined template (e.g. `student_invoice_copy` , `password_reset_by_admin` , `lead.application_link.sms`). When a piece of code wants to send a message, it just provides the template name + the dynamic values (student name, amount, date, etc.). The template fills the values in and delivers.

Provider routing. - **Email:** If a template is registered on MSG91 with the same name, it goes through MSG91 (from `mail.jdinstitute.com`). Otherwise it falls back to Gmail SMTP. This means we can move templates onto MSG91 one at a time, with no code change required. - **SMS:** Always goes through MSG91's DLT-approved flow (BulkSMS is kept available as a backup).

Audit trail. Every send (success or failure) is recorded with: who, when, what template, what response from the gateway, and what variables were used. The dispatch log is visible to admins for troubleshooting.

Recent improvements made this session: - Switched SMS from BulkSMS to MSG91 (BulkSMS was being blocked by our hosting provider's firewall). - Bypassed Cloudflare's bot detection that was blocking our requests. - Added MSG91 email routing for: admin password reset, student portal credentials. - Fixed sender display name so emails arrive as "JD Communications" instead of "admin.a".

5. Security & Compliance

Area	Status
Login required for all staff & student pages	✓
Role-based access (only authorised roles can do specific actions)	✓
Password rules (minimum 8 characters, hashing per industry standard)	✓
Brute-force lockout (auto-block after 5 failed login attempts)	✓
Activity audit log (every action by every user is recorded)	✓
File-type validation on uploads (no risky file types)	✓
Per-campus data isolation (staff only see leads/students from their campus)	✓
DLT-compliant SMS (Indian regulatory requirement)	✓
Forgot-password email enumeration protection (we never reveal whether an email is registered)	✓

6. What's Currently In Progress

Item	Status
MSG91 template verification — <code>password_reset_by_admin</code> , <code>student_portal_credentials</code>	Pending MSG91's internal review (templates created, waiting for approval)
MSG91 SMS template verification — <code>lead.application_link.sms</code> , <code>lead.fee_link.sms</code>	Pending DLT operator approval

These are workflow gates on MSG91's side — no code work is left. Once approved, the corresponding messages will begin sending without any further changes.

7. Hosting & Availability

- **Hosting:** PythonAnywhere (USA-based, India-friendly latency).
- **Domain:** `mahirsodagar.pythonanywhere.com` (will be moved to your own domain on go-live).
- **Database:** SQLite (built-in, no separate database server needed at current scale).
- **Backup strategy:** Daily snapshots + on-demand backups.
- **Uptime:** Hosted on a managed platform — uptime is the hosting provider's responsibility.

8. Summary

Area	Coverage
Modules fully built and working	12
Modules pending only third-party approval	0 (all done; only template verification on MSG91 pending)
Modules not yet started	None — every module from the original PHP system is covered.

The application is feature-complete for the original scope. From here, the work is mostly fine-tuning, training the staff, and handling any small adjustments that emerge from real usage.

Generated from the live codebase — every "Done" item above corresponds to working code currently deployed.